

Joseph Brazina

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EDUCATION

Virginia Polytechnic Institute and State University | Blacksburg, VA

M.Eng. in Industrial & Systems Engineering (Human Factors Engineering & Ergonomics) | *Expected May 2028*

James Madison University | Harrisonburg, VA

B.S. in Psychology, Minor in Sociology | *Graduated May 2026*

GPA: 3.50 | Major GPA: 3.75

RESEARCH EXPERIENCE

Research Assistant | Cognitive Eng. for Novel Technologies Lab, Virginia Tech | Sep 2026 - Present

- Supporting a study on whether expertise-tailored AR overlays improve bicycle inspection performance
- Preparing to assist with experimental-trial setup, HoloLens 2 calibration, data collection, and protocol adherence

Principal Investigator | Human Development Lab, James Madison University | Jan 2025 - May 2026

- Designed and led a 200+ participant IRB-approved study on sleep, mental health, and cognitive performance
- Developed and administered 3 Qualtrics surveys to collect study data across all three constructs
- Analyzed data from 200+ participants in SPSS using correlation, regression, and other related analyses
- Interpreted statistical findings to identify behavioral predictors of cognitive performance
- Presented study findings at the 2025 and 2026 JMU Psychology Symposium

Research Assistant | Auditory Perception Lab, James Madison University | Aug 2024 - Dec 2024

- Investigated categorical perception and working memory under conditions of cognitive load
- Designed an experimental study examining the effects of music on backward digit-span performance
- Administered 20+ participant sessions of computer-based auditory-perception tasks
- Configured 43 DirectRT experimental-trial files to implement task conditions and stimulus manipulations

ACADEMIC PROJECTS

App Usability Evaluation | Self-Directed Project | Fall 2026

- Conducted task analysis and heuristic evaluation of a mobile app's task flow, identifying usability issues
- Synthesized task-level risk points with heuristic findings to inform actionable recommendations
- Designed Figma mockups translating each recommendation into a proposed UI redesign
- Planning a moderated usability test to validate findings against real user behavior

Signal Detection Experiment | Human Information Processing Course | Fall 2026

- Led experimental design and implementation of a traffic-signal detection task
- Designed 80 stimuli manipulating glance time (200-516.7 ms) and deutan deficiency severity (24-100%)
- Analyzed signal-detection performance in JASP using response accuracy, false-alarm rates, and d'

Nintendo Switch 2 Redesign | Human Factors System Design Course | Fall 2026

- Lead a 3-person team evaluating the Nintendo Switch 2 home-screen user interface
- Define target users, user needs, and human factors deficiencies through usability testing and Figma prototype

PROFESSIONAL EXPERIENCE

Behavior Technician | ABA Today, Haymarket, VA | May 2025 - Aug 2025

- Implemented structured behavioral protocols across a caseload of 10+ clients
- 25+ client goals tracked via real-time performance data collection within a single session
- Conducted behavioral observations and adjusted task demands based on client responses
- Communicated client progress to clients' parents and Board Certified Behavior Analysts on a weekly basis

TECHNICAL SKILLS

Experimental & Data Methods: experimental design, human-subjects research, SPSS, DirectRT, statistics, correlation, regression, ANOVA, mediation analysis, t-tests, Excel, JASP

HF/UX & Tools: Figma, UI/UX prototyping, heuristic evaluation, hierarchical task analysis, stimulus design, Qualtrics

HONORS & PROFESSIONAL ACTIVITIES

Human Factors & Ergonomics Society | Institute for Industrial & Systems Engineers | Center for HCI | Psi Chi |

National Defense Society | JMU Psychology Research Symposium (2025, 2026) | Cum Laude Honors